

Arduino Nano — Setup and Code Upload (Step by Step)

This file provides a complete, easy, and practical guide for Arduino Nano (ATmega328) — from installing the IDE/drivers to uploading your first code, using the Serial Monitor, and troubleshooting common issues.

1. Quick Introduction / Requirements

Hardware:

- Arduino Nano (typically with Mini-USB connector)
- USB Mini-B cable (data supported)
- PC (Windows / macOS / Linux)

Software:

- Arduino IDE
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2. Install Arduino IDE

Windows

1. Download Arduino IDE from the [official Arduino website](#).
2. Run the installer and select "Install USB driver" if prompted.
3. Connect Nano with USB cable — Windows should detect a new COM port (check in Device Manager → Ports (COM & LPT)).

macOS

1. Download Arduino IDE and move it to Applications.
2. If your Nano uses CH340, install the CH340 driver (sometimes macOS security settings require manual "Allow").

Linux (Ubuntu/Debian)

- Extract official Arduino IDE archive and run the `arduino` executable, or install via:

```
sudo apt install arduino
```

- If you get a USB permission error, add your user to the dialout group:

```
sudo usermod -a -G dialout $USER
```

Then log out and log in again.

3. USB Driver (if COM Port Not Detected)

- Many clone Nanos use **CH340/CH341** USB-to-serial chips → install CH340 driver.
- FTDI-based Nanos may need FTDI drivers (often included in OS).
- Make sure you are using a proper data cable (not a charge-only cable).

4. Arduino IDE Settings (First Time)

1. Open Arduino IDE.
2. Go to **Tools → Board → Arduino Nano**.
3. Under **Tools → Processor**, select either **ATmega328P** or **ATmega328P (Old Bootloader)**. If upload fails, switch between them (many clones use old bootloader).
4. Go to **Tools → Port** and select the correct COM port (Windows: COM3/COM4, Linux: /dev/ttyUSB0 or /dev/ttyACM0).
5. For Serial Monitor, select the same baud rate as in your code (usually 9600 or 115200).

5. First Program — Blink

1. Open **File → Examples → 01.Basics → Blink**.
2. Example code:

```
void setup() {  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH);  
  delay(1000);  
  digitalWrite(LED_BUILTIN, LOW);  
  delay(1000);  
}
```

1. Verify Board/Port settings, then click **Upload** (right arrow button).
2. Onboard LED should start blinking every second.

6. Using Serial Monitor

- Add `Serial.begin(9600);` inside `setup()`.
- Use `Serial.print()` or `Serial.println()` to send data.
- Open `Tools → Serial Monitor` and select the correct baud rate.

Example:

```
void setup() {  
  Serial.begin(9600);  
}  
  
void loop() {  
  Serial.println("Hello from Nano");  
  delay(1000);  
}
```

7. Installing Libraries

- Open `Sketch → Include Library → Manage Libraries...`
- Search for the required library (e.g., DHT sensor, Adafruit_Sensor) and click Install.
- Use it in code with `#include <LibraryName.h>`.

8. Common Troubleshooting

- **Upload Error:** Try switching between “ATmega328P” and “ATmega328P (Old Bootloader)”.
- **Board not detected:** Check USB cable, install correct CH340/FTDI driver.
- **Permission error (Linux):** Add user to `dialout` group.
- **Code not running:** Check wiring, ensure correct COM port, verify power supply.

Now your Arduino Nano is ready for coding, uploading, and experimenting!